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Designing for AI Adoption in an Enterprise Environment

UX Factors Influencing Adoption of LLM-Based AI Systems

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This study investigates which UX factors are most important for supporting the adoption of LLM-based AI systems among domain experts with limited AI knowledge in an enterprise environment. Through a literature review, semi-structured interviews, prototype exploration, and evaluation, five key factors were identified: reliability, transparency, ease of use, usefulness, and perceived capability. The findings suggest that AI adoption in enterprise contexts depends not only on technical performance, but also on whether users can trust, understand, and effectively integrate the system into their daily work practices. In the studied context at Ericsson, reliability and transparency emerged as particularly important due to the high-stakes nature of contract-related work, where inaccurate outputs could have significant consequences. The study also identified perceived capability as an important factor, highlighting how users' beliefs about the maturity of AI technology influence their willingness to rely on it. Based on these findings, a prompt analyzer prototype was developed to explore how design interventions could support the identified UX factors. The evaluation showed positive effects on several of the factors, suggesting that user-centered design can play an important role in supporting AI adoption in enterprise environments.

SAMMANFATTNING

Denna studie undersöker vilka UX-faktorer som är viktigast för att stödja adoptionen av LLM-baserade AI-system bland domänexperter med begränsad AI-kunskap i en företagsmiljö. Genom en litteraturstudie, semistrukturerade intervjuer, prototyputforskning och utvärdering identifierades fem centrala faktorer: tillförlitlighet, transparens, användarvänlighet, användbarhet och upplevd kapabilitet. Resultaten visar att AI-adoption i företagskontexter inte enbart beror på teknisk prestanda, utan också på om användare kan lita på, förstå och effektivt integrera systemet i sina dagliga arbetsprocesser. I den studerade kontexten på Ericsson framträdde tillförlitlighet och transparens som särskilt viktiga faktorer på grund av den högkritiska karaktären hos kontraktsrelaterat arbete, där felaktiga resultat kan få betydande konsekvenser. Studien identifierade även upplevd kapabilitet som en viktig faktor, vilket belyser hur användares uppfattningar om AI-teknologins mognad påverkar deras vilja att förlita sig på den. Baserat på dessa resultat utvecklades en prototyp av en prompt analyzer för att undersöka hur designinterventioner kan stödja de identifierade UX-faktorerna. Utvärderingen visade positiva effekter på flera av faktorerna, vilket tyder på att användarcentrerad design kan spela en viktig roll för att stödja AI-adoption i företagsmiljöer.

CCS Concepts: • **Human-centered computing** → **User centered design**; • **Computing methodologies** → **Artificial intelligenc**.

Keywords: User Experience (UX), Artificial Intelligence (AI), AI Adoption, Large Language Model (LLMs)

Nyckelord: Användarupplevelse (UX), Artificiell intelligens (AI), AI-adoption, Stora språkmodeller (LLMs)

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1 INTRODUCTION

In recent years, artificial intelligence (AI) has become prominent across various sectors of modern society, influencing economic activities, social practices, and technological development [3, 27]. As these advances have accelerated, AI has become increasingly integrated into daily workflows, shaping how information is processed, decisions are made, and tasks are executed in organizational contexts [26].

In enterprise environments, AI is often promoted as a strategic resource for development, efficiency enhancement, and cost reduction [26]. While early applications primarily focused on automating repetitive and routine tasks, modern applications now play a broader role in professional contexts that extends beyond automation alone. Instead, such systems increasingly function as decision-support tools, providing analytical insights that help employees manage complex, information-intensive tasks and support informed decision-making [8, 38]. However, although the potential benefits of AI are widely recognized, many organizations face challenges in adopting and integrating these technologies into practice. The use of AI cannot be understood solely as process optimization, rather it introduces new ways of working, reshapes structures, enables new services, and alters the way value is created [18]. As these systems become embedded in everyday work practices, they interact with existing routines, roles, and user expectations, making adoption a broader organizational challenge rather than a purely technical one.

As a result, growing attention has been directed towards understanding the conditions that enable successful AI adoption. Existing research emphasizes that adoption depends not only on system performance, but also on a range of contextual and human-centered factors [21, 24, 29, 33]. One such factor is user experience (UX), as AI systems need to be designed with a clear understanding of user needs and interaction contexts to support user acceptance. Although research on UX frameworks for AI systems has increased in recent years, the overall body of work remains limited [13].

In addition, previous studies show that user experiences with AI systems are not uniform as perceptions and acceptance of new technologies depend on users' prior experience, digital skills, and cultural and demographic factors [31]. Consequently, user experiences with AI systems may differ significantly between user groups and usage contexts [12].

Research in the field of information systems further suggests that people with limited experience in a technical system are more likely to encounter uncertainty and computer anxiety, which can reduce perceived usefulness and negatively affect their intention to adopt new technologies [34]. Taken together, these findings highlight the need for research on which UX factors are perceived as important for AI adoption, particularly among users with limited AI knowledge within specific contexts and environments.

1.1 Research Context and Question

This identified need is reflected at Ericsson, which seeks to increase the adoption of an internal LLM-based AI solution. While earlier development efforts primarily emphasized technical capabilities, the organization has increasingly recognized the importance of user-centered design to support effective adoption and integration into employees' daily work. The AI solution is intended to support the sourcing department in contract-related tasks, such as contract analysis and interpretation, as well as the use of spend data to inform contract creation and negotiation, activities characterized by high complexity and critical decision-making.

In this context, many intended users are domain experts but have little prior AI knowledge. As a result, successful adoption depends not only on what the system can do, but also on how it is experienced,

understood, and trusted in practice. To address this need, this study focuses on identifying the UX factors that are most important when designing to support the adoption of an LLM-based AI system in enterprise work settings, with particular attention to domain experts with limited AI knowledge. Consequently, the study is guided by the following research question:

RQ: *"Which UX factors are most important when designing to support the adoption of an LLM-based AI system among domain experts with limited AI knowledge in an enterprise environment?"*

2 BACKGROUND

To understand which factors are important to support AI adoption, it is necessary to draw on previous research. Prior studies identify a range of factors influencing technology and AI adoption across different industries, user groups, and contexts. Many of these factors are conceptualized as broad, umbrella-like constructs shaped by multiple underlying dimensions, with trust being one such construct.

Rather than treating perceived trust as a single construct, this study adopts a more granular approach by identifying specific UX factors that have been shown to contribute to broader adoption-related constructs. This approach allows for a more precise examination of how different aspects of the user experience influence adoption in practice. Although these factors have been identified across a variety of domains, user groups, and settings, they are not assumed to be universally applicable. Instead, they serve as a theoretically grounded starting point for exploring which factors are perceived as most critical in the specific context of this study.

To identify and synthesize these UX factors, a literature review was conducted, the process of which is described in the methodology section 3.1. The review served two purposes: it established the theoretical background for the study, and it provided the initial conceptual framework that guided the empirical phases. For this reason, the identified factors are presented in this background rather than in results.

2.1 UX Factors Identified from Prior Research

The following sections describe each of the UX factors identified through the literature review.

2.1.1 Usefulness. Usefulness, in the context of technology adoption, refers to the extent to which an individual believes that using a particular system enhances job performance or supports the achievement of intended goals [5]. Previous research consistently identifies perceived usefulness as one of the most influential factors in determining whether users adopt or reject new technologies [5, 22]. Rather than capturing a single aspect, perceived usefulness often reflects multiple benefits, such as improved quality, increased efficiency, higher productivity, and effectiveness [4, 5]. When users recognize a clear and positive link between technology use and performance outcomes, their motivation to engage with and adopt the technology increases. Furthermore, qualitative studies indicate that attributes like high efficiency and rapid task execution are among the most frequently cited features contributing to the perceived usefulness of AI assistants, particularly in collaborative and co-creative settings [20].

2.1.2 Ease of Use. How much effort a system requires is central to whether users adopt it. Perceived ease of use is defined as the degree to which a person believes that a system can be used without significant effort [5]. In addition to perceived usefulness, research consistently shows that perceived ease of use is a key influential factor in the adoption of new technologies [14]. This influence is rooted in a basic cognitive tendency, as users naturally prefer technologies that require minimal effort over those

perceived as complex or difficult to operate. Consequently, systems that are seen to be easy to use are more likely to be accepted and adopted [17].

The relationship between perceived ease of use and technology adoption is often mediated by user' attitudes, as ease of use fosters positive perceptions that strengthen behavioral intentions [25]. This effect is particularly relevant for AI technologies, which are sometimes perceived as complex and difficult to use. Therefore, making AI-based consumer technologies intuitive to use is essential to promote positive adoption [4].

2.1.3 Transparency. Transparency refers to the extent to which users believe that a technological system communicates its purpose and processes in an open, clear and understandable way. Transparency is often conceptualized across two dimensions: informing users about *what* a system does and *how* it does it [43]. This factor has gained particular importance in the context of artificial intelligence, as it directly addresses the "black box" effect [2, 10]. The "black box" refers to situations in which AI systems produce outputs without providing users with information on the underlying logic or data. As a result, users may find it difficult to understand and evaluate how the system works [11]. Therefore, transparency plays a crucial role in users' willingness to adopt and continue using a system, as it can improve perceptions of trustworthiness. However, transparency must be carefully designed, since it can both strengthen and undermine users' trust. Providing too much information can erode trust by confusing users or causing information overload, highlighting the need for a cautious and balanced approach [16].

2.1.4 Controllability. When users feel they can influence and manage a system's outputs, they are more likely to engage with it. Perceived controllability refers to the degree to which users feel able to influence and manage system outputs. It reflects users' sense of self-efficacy and autonomy when interacting with a system in a given context and is considered an important factor in technology acceptance [19, 40]. Previous research shows that increased user control in human-AI collaboration leads to more positive user perceptions and higher engagement [36]. This includes allowing users to adjust system recommendations and provide input that the system can process. Studies further show that this sense of control influences users' trust in AI applications, which in turn affects the adoption intentions [39]. Beyond this, perceived controllability may also influence how safe users feel when interacting with AI systems. When users are able to influence system behavior, they are more likely to feel that they have control over what happens during interaction. This sense of control can contribute to psychological safety, meaning that users feel more comfortable, secure, and less uncertain when engaging with the system [39].

2.1.5 Reliability. Perceived reliability refers to the belief that a technology functions correctly, consistently, and as intended [4]. It is closely related to trust, as it reflects the consistency and integrity of a system's operation. In AI technologies, reliable performance through accurate and consistent information delivery is essential to foster user trust and reinforce confidence in the system [4, 30]. Previous research further emphasizes that perceived reliability is an important factor influencing technology acceptance and adoption [15, 32]. When perceived reliability is too low, users may be less likely to adopt or rely on the system and may ignore its recommendations, thereby missing out on the potential advantages of the technology. In contrast, when users overestimate the reliability of an AI system relative to its actual performance, they may neglect its potential for error, resulting in overconfidence and potentially poor decision-making [35].

2.1.6 Fairness. Perceived fairness refers to an individual's subjective judgment of whether a decision, process, or outcome is fair and unbiased. Unlike formal or mathematical definitions of fairness, perceived fairness is shaped by personal intuition, emotional responses and the surrounding context [6]. Prior research indicates that explanations in AI systems are more likely to be perceived as fair when they are considered reasonable, understandable, and responsive to the needs and concerns of those affected by the decision [23]. Findings suggest that users are less likely to adopt an AI system when they feel its outputs are unfair or biased toward certain outcomes. Such perceptions of unfairness weakens trust, which in turn reduces the likelihood of adoption [37, 42]. This underscores that fairness in AI systems cannot be fully addressed through technical solutions alone, as it also depends on how users experience and interpret system outputs [28].

2.1.7 Risks. Perceived risks refer to users' subjective judgements of the potential negative consequences of adopting a technology and have been consistently identified as an important factor in related studies [9]. Research on technology acceptance and resistance shows that adoption decisions are not based solely on perceived benefits, but rather on how these benefits are weighed against associated risks [1]. When perceived risk is high, people are significantly less willing to adopt or continue using such technologies [7]. This effect is particularly pronounced in high-stakes environments, where multiple types of risk must be considered [41]. Prior research identifies six types of perceived risks; performance, financial, social, psychological, safety and opportunity/time. Overall, these risk perceptions play a central role in shaping adoption decisions [20].

3 METHOD

This study adopts a qualitative and exploratory research approach, which is appropriate given the aim of understanding how non-technical users experience and evaluate AI systems in their work context. Since user perceptions, expectations, and interpretations of AI-supported work are complex and context-dependent, qualitative methods are well suited to capture nuanced experiences that cannot be easily quantified. The research combines a literature review, semi-structured interviews, thematic analysis, and prototype-based exploration to iteratively identify, refine, and explore UX factors related to AI adoption.

3.1 Initial UX Factor Identification

This section describes the process used to identify an initial set of UX factors through a review of prior literature. The factors identified through this process are presented and described in the background section.

3.1.1 Purpose of the Literature Review. The literature review was conducted to identify and define an initial set of UX factors relevant to the adoption of LLM-based AI systems in enterprise environments. It provided the theoretical foundation for the empirical phases of the study and established a conceptual framework for understanding user experiences with such systems.

The review drew on research within user experience, AI adoption, and broader technology acceptance literature. By synthesizing prior work, it identified experiential aspects that had been shown to influence technology acceptance and adoption in AI contexts. The resulting UX factor list served as an initial analytical framework for the interview phase and remained open to refinement based on empirical findings.

3.1.2 Definition of UX Factors. In this study, UX factors are defined as recurring aspects of the user experience that influence how users perceive, interpret, and engage with the systems. These aspects may relate to cognitive, emotional, behavioral, or contextual dimensions of interaction. UX factors describe what matters in the user experience, rather than specific interface features or technical capabilities.

For example, *perceived transparency* is a UX factor because it describes a quality of the user experience, the extent to which users feel they understand how the system works and why it produces certain outputs. In contrast, *showing confidence scores* is a design feature that might support transparency, but is not itself a UX factor.

3.1.3 Criteria for UX Factor Identification. To be identified as a UX factor in this study, a concept had to meet following criteria:

Experiential level. The factor had to operate at the level of user experience rather than referring to technical features, interface elements, or implementation methods. It needed to capture how users subjectively perceive, feel, or relate to the system.

User-centered focus. The factor had to reflect the perspective of individual users, rather than organizational policies, governance structures, or system-level performance metrics.

Multi-source validation. To be included in the initial factor list, a concept had to be grounded in multiple independent sources, ensuring it represented a consistently recognized experiential dimension rather than an isolated finding.

Analytical utility. The factor had to be formulated concretely enough to serve as an analytical lens during empirical data collection and analysis, while remaining broad enough to capture varied manifestations across different user contexts.

These criteria guided both the selection of relevant literature and the extraction and synthesis of UX factors from that literature.

3.1.4 Analysis and Synthesis Procedure. The literature analysis followed an iterative approach in which sources were identified, reviewed, and analyzed for UX-related insights. Each source was read with a focus on descriptions of user experience, paying particular attention to how users feel, perceive, understand, or relate to AI systems, including expressions of uncertainty, understanding, trust, and perceived control.

Relevant passages from the reviewed literature were systematically documented in a spreadsheet to organize and compare findings across sources. This systematic documentation ensured traceability and enabled comparison across sources to identify recurring themes. The literature review was considered complete when additional sources reinforced existing UX factors using different terminology rather than bringing in new insights. At this point, each factor was supported by multiple sources and accompanied by clear working definitions. Final confirmation was obtained in consultation with supervisors at KTH.

3.2 Semi-Structured Interviews

This section describes the semi-structured interviews conducted to explore user experiences with LLM-based AI systems. The interviews were used to validate and refine the UX factors from the literature review and identify additional insights relevant to the specific setting.

3.2.1 Purpose of the Interview Study. The interview study was conducted to empirically investigate how domain experts with limited AI knowledge experience and evaluate LLM-based AI systems in the context of contract-related work. While the literature review provided an initial theoretical framework, the interviews were essential for grounding these factors in the specific professional context of the study. The interviews served two purposes: to validate and refine the initial set of literature-derived UX factors and to identify additional experiential dimensions that may not have been sufficiently captured in prior research, particularly aspects specific to this user group and this organizational context.

3.2.2 Design of the Interview Study. Semi-structured interviews were chosen as the primary method because they combine systematic thematic coverage with the flexibility to explore participants' perspectives in depth. This design is particularly suited to studying user experience, as perceptions of AI systems are context-dependent and difficult to predict in advance. The semi-structured format allowed the interview guide to serve as an analytical framework based on the literature-derived UX factors, while also leaving space for participants to surface additional concerns, priorities, or experiential dimensions. To support the planning and facilitation of the interviews, a structured overview of the interview guide was developed in Figma. This visual map of the thematic areas and question flow served as a practical reference during sessions. In practice, however, the guide functioned more as a safety net than a strict script: the majority of participants engaged extensively and proactively with the topics, moving fluidly between themes based on their own experiences and priorities. Questions were primarily used to assess which relevant factors had already been addressed and which required further exploration, or to gently introduce topics that had not emerged organically in the conversation.

3.2.3 Participant Selection. A total of twelve interviews were conducted with employees at Ericsson, all working within the sourcing function but across different sub-processes. The sample was purposely selected to reflect the profile of intended users of the AI system: experienced professionals with deep domain knowledge but limited familiarity with AI. Participants were predominantly in the 40–60 age range and held senior positions in their respective areas. This focus was motivated by the study's interest in how established professional expertise and expectations shape perceptions of AI-assisted work.

The sample was diverse in several respects. It included six women and six men, and interviews were conducted in both Swedish and English, reflecting Ericsson's character as a global organization where employees collaborate across regional offices. Participants also varied in their current attitudes toward AI: some held a critical or skeptical stance, questioning the relevance or reliability of AI tools in professional settings, while others had already begun experimenting with AI in parts of their work. This variation was intentional, as it enabled the study to capture a broader spectrum of user experiences and attitudes among participants.

3.2.4 Interview Procedure. Each interview was conducted individually and lasted between approximately 50 and 100 minutes. All sessions were recorded with the informed consent of the participants. Following each session, recordings were transcribed using Microsoft Copilot, after which the transcripts were reviewed and corrected for accuracy.

The interviews were conducted in a conversational manner, where participants frequently introduced topics and concerns unprompted, often drawing on concrete work experiences to illustrate how they anticipated or had already encountered AI in their professional roles. This organic dialogue was treated as methodologically valuable, as it allowed participants to prioritize and articulate what they considered most meaningful in their own terms rather than responding only to predefined questions. Throughout the

interviews, an open and exploratory stance was maintained in order to capture experiential dimensions that emerged from the discussions.

3.2.5 Ethical Considerations. Prior to each interview, participants were informed about the purpose of the study, that their participation was entirely optional, their right to withdraw at any time without consequence, and how the collected data would be used and stored. No personally identifiable information is reported in this thesis. All recordings and transcripts were handled and stored securely throughout the research process.

3.3 Prototype-Based Exploration

While the interviews served to validate and refine the identified UX factors, they also surfaced a recurring practical challenge that pointed toward a concrete area for design exploration. This led to a complementary prototyping phase, aimed at investigating how specific design interventions might support these UX factors and address experiential barriers to adoption.

3.3.1 Purpose of the Prototype. Following the interview phase, it became clear that many participants experienced difficulties in formulating effective prompts when interacting with LLM-based AI systems. Given that the quality of an AI system's output is highly dependent on the quality of the user's input, prompting emerged as a critical aspect of the user experience and could, in turn, affect several of the UX factors identified earlier in the study. This observation motivated an exploratory prototyping phase aimed at investigating how prompt-assisting design features might support these UX factors and, ultimately, support AI adoption.

Within Ericsson's internal AI solution, several specialized agents exist for different purposes. One of these is the spend agent, which supports tasks related to spend data and analysis. In this context, effective prompting is especially important, since users often need outputs that accurately reflect specific figures, constraints, or analytical needs. If key details are not included in the prompt, the AI may make implicit assumptions regarding, for example, time frame, currency, or other contextual parameters. This can lead to outputs that are less relevant or less trustworthy from the user's perspective. Several interview participants explicitly expressed a need for support in how to formulate prompts, making this a particularly relevant area for design exploration.

3.3.2 Prototyping Process. Given the interactive and iterative nature of prompt-assist functionality, it was considered appropriate to move directly into development rather than relying solely on static sketches or wireframes. To maintain an open and exploratory approach, the ideation phase was intentionally initiated before reviewing existing solutions, avoiding early influence from established design patterns. First concepts were therefore sketched freely to generate a broad range of ideas that could later be evaluated against both user needs and technical constraints to determine which directions to pursue further. These concepts were subsequently implemented directly within the AI system's development environment, allowing design decisions to be assessed in the context of real system behavior rather than in isolation.

The prototype was developed iteratively using Claude Code as a development tool, alongside continuous input and technical guidance from a supervisor at Ericsson. Each iteration refined both the interaction design and the underlying logic of the prompt-assist features, with adjustments made in response to practical constraints encountered during development. The prototype went through multiple cycles

of revision, with each version informing the next based on observed limitations and emerging design opportunities.

The aim of the prototyping phase was to explore how different design choices might support the UX factors identified in the study. The resulting prototype therefore reflects a combination of design exploration and technical feasibility within the given constraints, and serves as a basis for the following evaluation.

3.4 Evaluating Prototype's Impact on UX Factors

3.4.1 Purpose of the Evaluation. The prototype evaluation was conducted to investigate how it was experienced by intended users and, more specifically, how it influenced the UX factors identified as most important for AI adoption in the earlier interview phase. While the prototype was developed as an exploratory response to the prompting difficulties surfaced during the initial interviews, its actual contribution to the user experience could only be assessed through hands-on testing with users. The evaluation therefore served two related purposes: to gather feedback on the prototype as a design artifact, and to examine whether and how the prompt-assist features affected users perceptions along the identified UX factors.

3.4.2 Design of the Evaluation. The evaluation combined a hands-on prototype testing session with a semi-structured interview. This design was chosen because the focus was not only on whether participants could operate the prototype, but on how they perceived and reasoned about it in relation to their own work practices. A purely observational usability test would not have captured the experiential and reflective dimensions central to the research question. The semi-structured format ensured that each session covered the same set of UX factors while leaving room for participants to bring up reactions, comparisons, and concerns that emerged during the interaction.

An interview guide was prepared in advance, with questions structured around the UX factors identified as most important in the initial interview phase. Questions were intentionally framed in an open manner, asking participants to reflect on how the prototype affected their willingness to rely on the system, their understanding of what the agent needed to perform well, the perceived effort required to interact with the agent, and the kinds of situations in which a feature like this would be valuable in their work.

3.4.3 Participant Selection. Five participants from the sourcing function at Ericsson were recruited for the evaluation. The sample was drawn from the same broader population as the initial interview study, in order to maintain consistency with the user group the prototype was designed for.

3.4.4 Evaluation Procedure. Each session lasted between approximately 20 and 30 minutes and was conducted individually. The session began with a brief introduction in which the motivation for the prototype was explained and the underlying interaction principle was demonstrated in a simple example. The participant was then invited to use the prototype themselves. The interview phase followed directly after the testing and was structured around the UX factors. All sessions were recorded with the informed consent of the participants and conducted in Swedish, reflecting the participants' working language. Recordings were transcribed and subsequently translated into English for the purposes of analysis and reporting.

3.4.5 Thematic Analysis. The transcribed material was analyzed thematically, using the identified UX factors as the primary analytical lens. For each factor, statements that described how the prototype affected the participant's perception or behavior were extracted and grouped. Particular attention was given to whether the prototype was experienced as supporting, weakening, or having no clear effect on a given UX factor, and to the reasoning participants offered for these perceptions. Quotations were selected to illustrate recurring patterns across the five participants.

4 RESULTS

This chapter presents the findings from each phase of the study. First, the final set of important UX factors identified through the qualitative interviews is presented. Second, the design and development of the prototype is described and finally, the results from the prototype evaluation.

4.1 Final Set of UX Factors

From the interviews, several UX factors were identified as important for increasing users' adoption of AI in an enterprise setting. Among the factors identified in the literature, four emerged as particularly important in the empirical material: reliability, transparency, ease of use and usefulness. In addition, the analysis revealed a new factor not explicitly emphasized in the theoretical framework, referred to here as perceived capability, which relates to users' beliefs about the overall capacity and maturity of the AI system. These factors are presented below as the most critical influences on adoption in the studied context.

4.1.1 Reliability. Perceived reliability emerged as one of the most critical factors influencing participants' willingness to rely on the AI system. Many interviewees emphasized that even a single incorrect output could significantly reduce their reliability in the system. In the context of contract-related decision-making, where accuracy is crucial, participants expressed a low tolerance for errors. Several respondents described reliability as a key determinant of continued use. When the system produces incorrect or inconsistent outputs, users may quickly abandon it. One participant explained the consequences of inaccurate data:

> *"If the data is wrong even once, it can cause a massive setback, because you can't trust what you see." - Participant 6*

Another participant described how inconsistent performance discourages further use:

> *"Sometimes it works, sometimes it doesn't. And if it doesn't work, then you tend to leave it pretty quickly nowadays." - Participant 8*

Similarly, another interviewee emphasized that basic errors strongly undermine confidence in the technology:

> *"If AI fails once on the most basic, trivial things as a human and an engineer you just think, 'What the hell – away with this." - Participant 5*

These findings suggest that perceived reliability plays a fundamental role in determining whether users continue engaging with the AI system. In high-stakes professional environments, reliability appears to function as a baseline requirement for adoption.

4.1.2 Transparency. Transparency was another factor consistently emphasized by participants. Many respondents highlighted the importance of being able to verify the information provided by the AI system,

particularly when the system generates recommendations or analytical outputs from data. Participants frequently associated transparency with the ability to trace sources, verify information, and understand how conclusions were reached. This ability was described as essential for evaluating the credibility of the system's outputs. One participant pointed out the importance of source attribution:

> *"Trustworthiness is quoting your sources right on the spot. If something is being said, or something needs to be reviewed or fact-checked, I need the ability to fact-check it right there." - Participant 7*

Similarly, another participant compared interactions with AI to interactions with employees:

> *"Just as I would question any of my employees about how they came to a conclusion, I'm going to demand the same from AI from a transparency standpoint." - Participant 1*

Several interviewees also expressed concerns about relying on outputs without understanding their origin:

> *"If you can't see how the AI came to its conclusions, then it has limited value." - Participant 3*

Another participant emphasized that references are necessary to avoid blind trust:

> *"AI needs to point to its references, because the dangerous part is when you blindly trust what the AI tells you and don't apply any kind of critical thinking." - Participant 5*

Transparency was therefore perceived as an important UX factor for enabling users to critically evaluate AI-generated information and understand the system's underlying reasoning.

4.1.3 Ease of use. Ease of use was frequently discussed in relation to the effort required to interact with the AI system. Participants emphasized that complex interaction processes or inefficient workflows could discourage usage. Several respondents explained that systems requiring extensive prompting or multiple interaction steps may reduce motivation to engage with the technology. One participant described how complex interaction paths can reduce usability:

> *"If you have to go through a marked path to get to the strategy, starting in one place and then following the whole route - no one is going to have the energy for that." - Participant 3*

Participants also expressed interest in interface features that simplify interaction. For example, guided interactions or structured prompts were perceived as helpful for users with limited experience. One participant described a positive experience with such functionality:

> *"That was the first time I saw it done in a way where the system asks a follow-up question before continuing. Instead of forcing the user to write a new prompt, it presents options like A,B,C." - Participant 4*

Similarly, another respondent noted that users lack expertise in prompt design:

> *"What I might be missing is that at my level I usually just ask the question I have. I'm not really trained in prompting, and I feel like I could probably get better answers if I were maybe with some guidance like, "Have you considered this when asking your question?"" - Participant 9*

These findings indicate that intuitive interaction design and support for users with limited AI experience may significantly enhance ease of use and thereby influence adoption.

4.1.4 Usefulness. Perceived usefulness also influenced participants' attitudes toward the AI system. However, interview responses indicated that usefulness was closely linked to both reliability and output quality. Some participants questioned whether the AI system provided meaningful value in its current state. In particular, concerns were raised about the accuracy and substance of generated outputs. One participant expressed concerns about the potential risks of inaccurate outputs:

> *"I think this one is completely useless, because if you use it, it could give you an answer that's wrong by a factor of ten, and that's dangerous." - Participant 3*

Another participant questioned the practical value of AI-generated text:

> *"AI is very good at generating a lot of words, but the context and the content often feel pretty empty to me. Personally, I don't really have that problem, I'm quite verbal and good at writing reports so to me it often just becomes a lot of words without much substance." - Participant 5*

These responses suggest that users evaluate usefulness not only in terms of efficiency, but also based on the quality and relevance of AI-generated information. At the same time, some participants described efforts to increase engagement with the system by demonstrating its potential benefits to colleagues. As one participant explained:

> *"I think at the beginning people were not very eager to try something new, so we had some one-on-one talks and showed them vivid cases to spark their interest." - Participant 12*

This indicates that perceptions of usefulness may develop over time as users become more familiar with the system and observe concrete examples of its potential applications.

4.1.5 Capability. In addition to the factors identified in the literature, the interviews revealed another important factor influencing AI adoption: perceived capability. This factor refers to users' beliefs about the technological maturity and overall capacity of the AI system. Participants often discussed AI in terms of what they believed the technology was currently capable of achieving. These perceptions influenced expectations regarding system performance and appropriate use cases. Some respondents described the technology as still being in an early stage of development:

> *"Because the reality is that this is a fledgling technology. It's still very immature and very incomplete, and you don't know whether the technology hasn't caught up to the question, or whether the person asking the question isn't doing a good job. From what I've found, most of the time it's the technology." - Participant 7*

Another participant compared the development of AI to a learning process:

> *"I think it's like this: it has to learn to crawl before it can walk and before it can run. The probability that it can run right now is incredibly low, so I just want it to crawl on the floor and pick up a few things for me, that's about the level I think it's capable of." - Participant 3*

These perceptions suggest that users may calibrate their expectations based on their beliefs about the maturity of AI technology. As a result, perceived capability may influence both how users evaluate system performance and how they decide when to use the system.

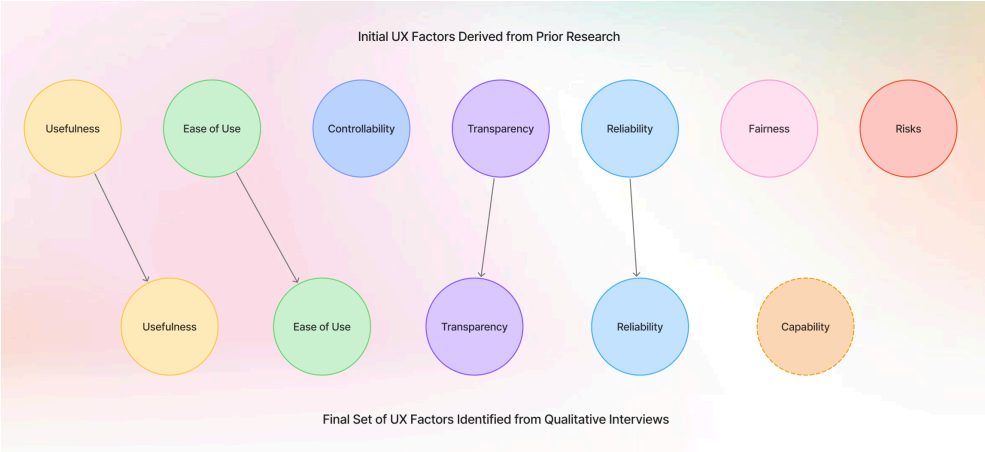


Figure 1: Initial UX Factors to Final set of UX Factors

4.2 Prototype: The Prompt Analyzer

Based on the UX factors identified in the interviews, a prototype was developed to explore how these factors could be supported through concrete design interventions. The following section presents the Prompt Analyzer, including its design rationale, implementation, and key design decisions.

4.2.1 Motivation for the Prompt Analyzer. The motivation for this design is derived from insights from the interviews and reflects a common interaction pattern observed in contemporary LLM-based systems. In a typical workflow, when users submit a poorly specified prompt, they receive either: a direct response where the system has made implicit assumptions due to missing information, or a response containing follow-up questions requesting clarification.

Both outcomes introduce potential challenges. When the system makes incorrect assumptions, users may receive inaccurate or misleading outputs. This can lead to increased costs in terms of time and effort, but more importantly, negatively impact perceived reliability, usefulness, and system capability, which were identified as critical UX factors in the study. On the other hand, responses in the form of follow-up questions may interrupt the user's workflow. Although such clarifications can be beneficial, users typically expect a direct answer, and being required to reformulate their input without anticipating or explicitly requesting it can introduce additional interaction effort.

To address these challenges, the prototype introduced a prompt analyzer, a feature designed to provide feedback on user prompts before they are processed by the AI system. The concept shifts part of the interaction from a post-response refinement process to a pre-response refinement process illustrated in Figure 2. Instead of reacting to poor outputs, users are supported in improving their inputs upfront.

4.2.2 Interface and Interaction Design. To provide context for the prototype, Figure 3 presents the interface of the AI system, specifically the view of the Spend Agent chat. This overview illustrates the environment in which the Prompt Analyzer is integrated.

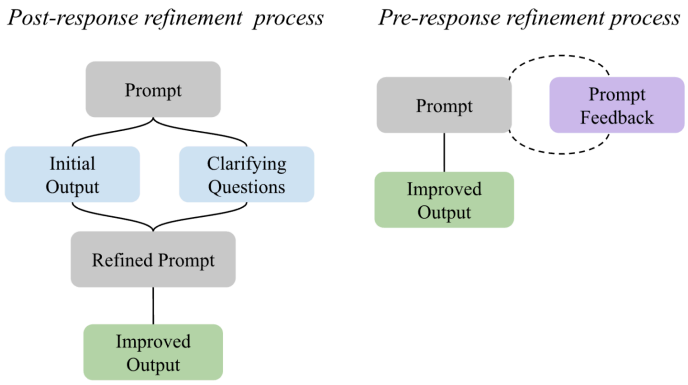


Figure 2: Interaction Flow Without and With Prompt Feedback

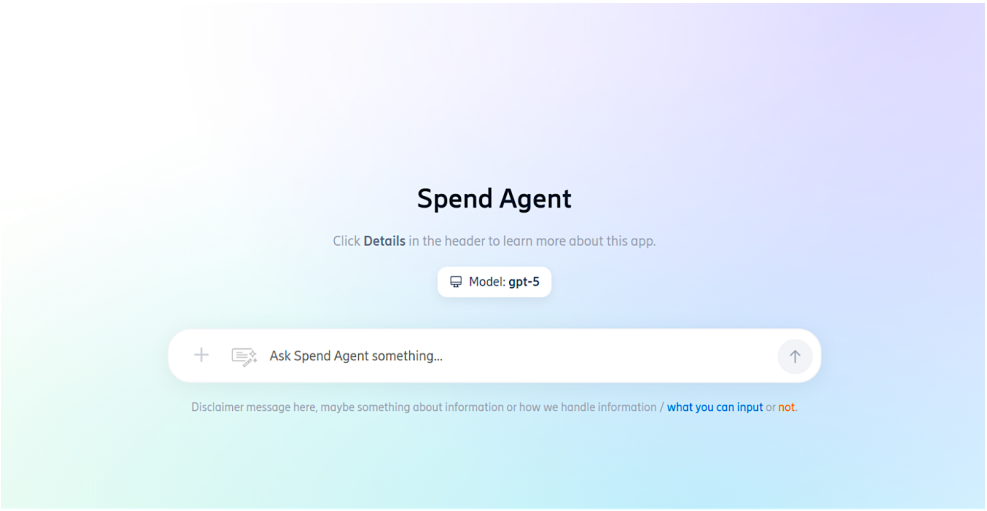


Figure 3: Spend Agent Chat Interface Overview

The Prompt Analyzer is accessed through a button placed next to the input field in the chat interface, which is visually indicated as active through a turquoise-colored icon once the user has typed a prompt. When clicked, the system analyzes the input and returns a set of suggestions, each targeting a specific aspect of the prompt that could be made more precise.

Each suggestion is displayed as a bubble containing three elements: a priority badge, a short description of the identified issue and a suggested improvement, and a row of clickable option pills representing concrete alternatives the user can choose from. Priority levels are color-coded to create a clear visual hierarchy: high priority suggestions are marked in red, medium in yellow, and low in blue.

To support prioritization of suggestions, common types of assumptions made by the AI system were first identified. This was done through systematic testing of the spend agent, where a range of incomplete prompts were submitted to observe how the system responded when key details were missing. The testing revealed four recurring dimensions where the agent consistently made implicit assumptions: time

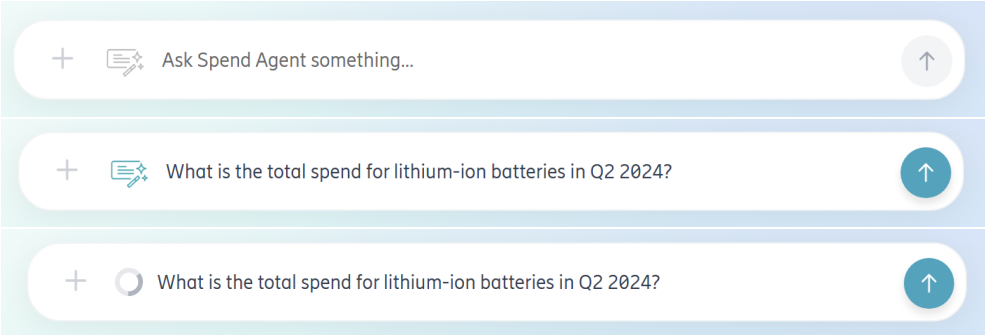


Figure 4: Prompt Analyzer Button Next To Input Field

window, currency, geographical scope, and category matching. These dimensions were therefore given particular attention in the analyzer's logic and treated as high priority, as assumptions in these areas risk producing outputs that do not reflect the user's actual intent.

Each suggestion includes multiple clickable option pills, typically four to five alternatives, allowing users to select the one that best matches their intent. This was an intentional design decision made after early versions presented only a single suggestion per category, which proved limiting when that option did not match what the user had in mind. By offering several alternatives, the design increases the likelihood that users can find a suitable option without manual input, while still allowing them to enter their own input if none of the options are appropriate.

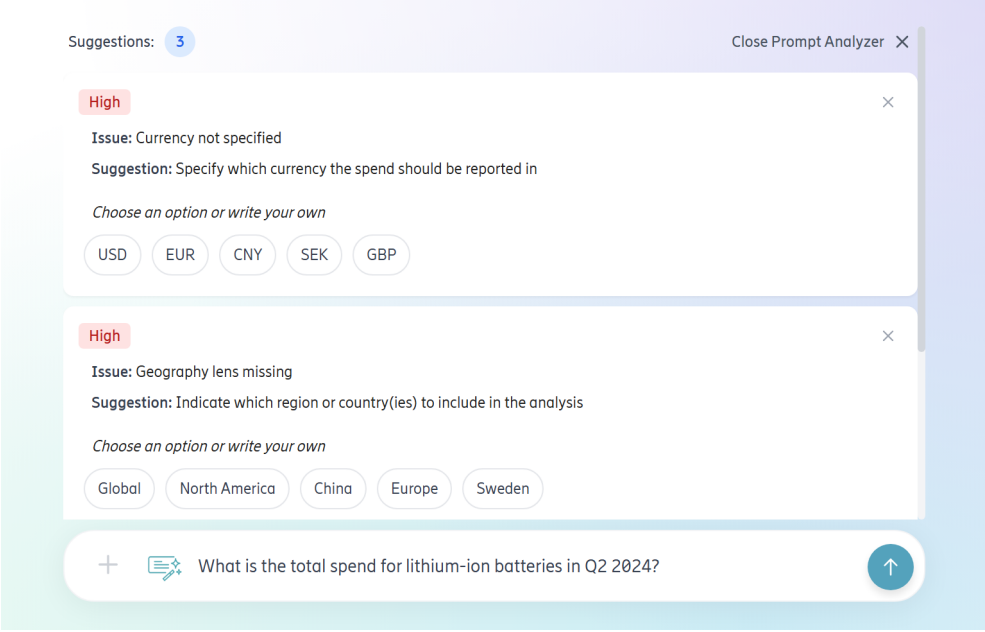


Figure 5: Prompt Analyzer Suggestions with Priority Badge and Option Pills

When a user selects an option, the prompt in the input field updates immediately to reflect the addition. This real-time update replaced an earlier iteration in which users first had to select all desired options

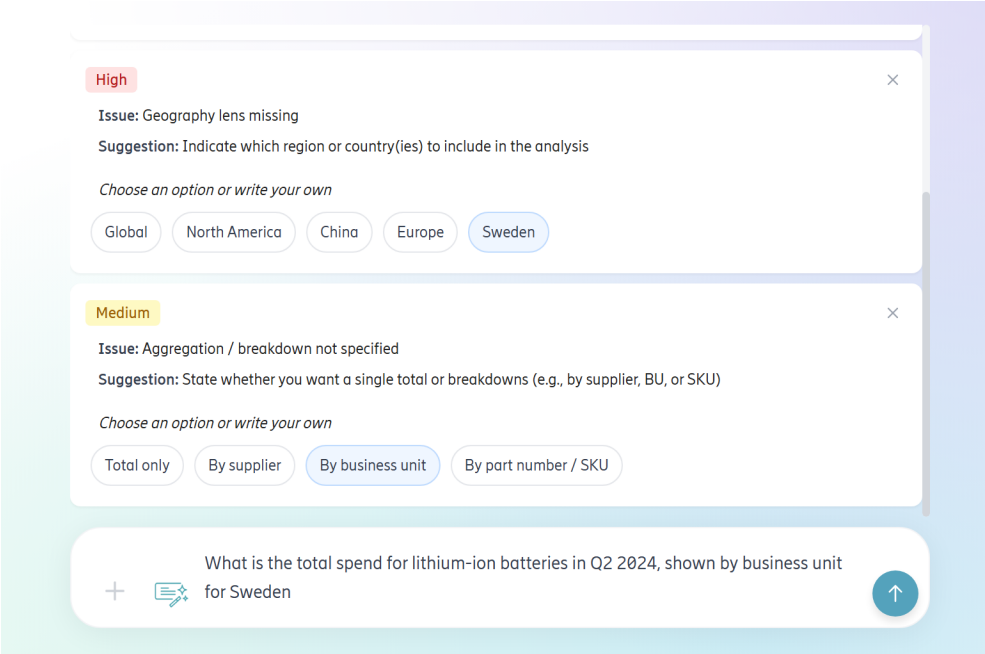


Figure 6: Selected Option Pills Added to the User's Prompt

across the suggestions and then click a separate apply button to generate the updated prompt. The earlier approach did not provide real-time feedback, as changes only became visible once all selections were confirmed at once. This made it harder to follow how the prompt evolved, whereas the updated design improved transparency and traceability of changes. If a user deselects an option, the prompt returns to a version without that addition.

Once the user has refined their prompt using the Prompt Analyzer, the final prompt can be submitted to the agent either by clicking the send arrow or pressing the Enter key. This enables users to directly proceed with a more structured and complete input, which increases the likelihood of receiving relevant responses without the need for additional clarification after submission.

4.2.3 Technical Implementation.

Suggestion Generation. The Prompt Analyzer operates by passing two inputs into a meta-prompt: the spend agent's system prompt, which defines its behavior and capabilities, and the user's original prompt. Before any analysis takes place, the meta-prompt performs a check to determine whether the input is actually a spend analysis question. If the user has typed something unrelated, such as a greeting or a question outside the scope of spend data, the system returns a single structured suggestion indicating that the input does not appear to be a relevant query. This step was considered important to prevent the analyzer from generating suggestions for inputs that were never intended for the spend agent in the first place.

If the input passes this check and is identified as a valid spend question, the meta-prompt proceeds to identify two to five specific improvement suggestions. Particular attention is given to the four most common missing dimensions identified during testing, though other types of suggestions are also generated depending on what is found to be missing or unclear in the prompt.

Prompt Reconstruction. One technically complex aspect of the prototype was ensuring that selected options were inserted into the prompt in a natural and grammatically coherent way. In early iterations, options were simply appended to the existing prompt, which produced unnatural and sometimes confusing phrasing, for example outputs like ”Show me spending Q1 2024 USD top 10 EMEA” with no connecting language. To address this, a reconstruction function was developed that categorizes each selected option based on both the content of the option itself and the context provided by the suggestion it came from. The function analyzes the option text for patterns such as year formats, quarter references, currency codes, or geographic terms, while simultaneously examining the issue and improvement text of the suggestion to understand what type of information the option represents. Through this dual analysis, each option is classified into one of eight semantic categories: time, time period, currency, geography, category, metric, filter, or grouping.

The reconstruction function operates on a critical design principle: it always rebuilds the entire prompt from the original user input rather than modifying the current state. Each time a user selects or deselects an option, the system returns to the original prompt and reapplies all currently selected options from scratch. This stateless approach proved essential during development, as early testing with incremental modifications led to compounding errors where one insertion would affect the placement of subsequent ones, resulting in duplicated content, incorrect ordering, and options appearing in unexpected positions. By rebuilding from the original each time, the function ensures consistent and predictable results regardless of the order in which users select options.

With this foundation, each categorized option is inserted using context-appropriate connecting language and positioned where it most naturally belongs in the sentence structure.For instance, time period modifiers like ”quarterly” or ”monthly” are inserted before existing time ranges to modify them, while currency options consistently use the phrasing ”in [currency]” and geographic specifications use ”for [region]”. The function also performs context-aware duplicate checking to avoid redundantly inserting information already present in the original prompt.

After all categorized insertions, a post-processing step normalizes spacing, removes duplicate conjunctions, and ensures proper capitalization, resulting in prompts that read as naturally constructed queries rather than assembled fragments.

4.2.4 *Iterations and Constraints.*

The prototype went through several notable iterations across both design and technical dimensions. The first major design decision concerned how the analysis should be triggered. An early concept involved real-time feedback that would appear automatically a few seconds after the user stopped typing, providing continuous guidance without requiring any additional action. After discussion with the supervisor from Ericsson, this approach was set aside due to technical constraints and the time available for the exploration. A manually triggered button was implemented instead, giving users control over when they wanted to analyze their prompt.

One limitation that was identified but not resolved within the scope of this exploration concerns the absence of a natural stopping point. Because the AI will always be able to find new potential improvements, the analyzer does not have a way of recognizing when a prompt is sufficiently complete. As the prompt becomes more detailed, the suggestions shift toward lower priority improvements, but they never stop entirely. Ideally, a success metric or saturation indicator would be developed to recognize when a prompt has reached a sufficient level of detail, and potentially provide positive feedback to the user at that point. This was not implemented due to time constraints but remains an interesting direction for future development.

4.3 Prototype's Impact on UX Factors

The prototype evaluation provided insight into how the prompt analyzer was experienced by intended users and how it impacted the UX factors. The findings are presented below, organized by the five UX factors identified in section 4.1.

4.3.1 Reliability. Reliability emerged as one of the UX factors most clearly affected by the prompt analyzer. Several participants described how the structure of the analyzer's feedback gave them a stronger sense that the system would be able to produce dependable answers, since the relevant information was being clarified upfront rather than implicitly assumed. One participant emphasized that the analyzer addressed a recurring source of uncertainty in interactions with the agent, namely not knowing what information the system actually needed to perform well:

> *"Yes, especially when it comes to ensuring that I provide the right kind of information that the agent needs to give reliable answers. Otherwise, it becomes a bit of a black box and I end up guessing what it needs." - Participant 2*

Another participant described how the act of being asked relevant clarifying questions, rather than receiving an immediate answer, increased the perceived trustworthiness of the eventual output:

> *"At the very least, it gives me the impression that it has understood what I'm writing and that it has some questions, which makes me trust the answer more when it finally comes." - Participant 1*

These reflections suggest that the prompt analyzer supports perceived reliability primarily by helping users provide the information the agent needs to generate accurate answers. By making clear which dimensions of the prompt are likely to influence the output, the feature reduces the risk of incomplete or misaligned input and, in turn, unreliable answers. Reliability is therefore built at the point of input rather than after the fact.

4.3.2 Transparency. Transparency was the factor most consistently associated with the prompt analyzer across participants. Participants described how the analyzer surfaced aspects of the underlying logic that would otherwise remain hidden, allowing them to see how different inputs were likely to shape the agent's output. One participant described this in terms of being able to follow the reasoning structure of the system:

> *"I can kind of see how it's thinking and what the information tree in the AI looks like, which helps me understand that these different choices lead to different spending." - Participant 3*

Another participant framed transparency more relationally, describing the analyzer as a counterpart that contributed to the joint formulation of the prompt rather than a passive tool:

> *"It's like you're not completely alone in describing exactly what you need - you get a kind of sounding board, someone contributing from the other side, which also increases transparency." - Participant 4*

This indicates that the prompt analyzer contributes to transparency by externalizing parts of the agent's reasoning that would otherwise remain hidden. Rather than only seeing the final output, users gain insight into how the agent is structured and how different inputs map to different outputs. In doing so, the feature appears to reduce the "black box" effect that participants described as a barrier to adoption in the initial interview phase.

4.3.3 Ease of Use. Ease of use was another factor where participants reported a clear positive effect. The analyzer was described as lowering the threshold for engaging with the agent, particularly in situations where users felt uncertain about how to formulate an effective prompt. One participant described how the prototype reduced the hesitation associated with writing the initial prompt and connected this to broader patterns of AI adoption among colleagues:

> *"I think it became easier to use the agent because you don't have to be so afraid of writing your initial prompt. You can just write about a topic you want an answer to, and then you get guidance. I think a lot of people are a bit held back by that - they don't feel skilled enough to write prompts, so they don't do it and instead keep doing things the way they always have with paper, pen, scissors, and tape." - Participant 1*

Another participant similarly emphasized that having pre-formulated suggestions reduced the cognitive effort of articulating one's own needs:

> *"It's hard to formulate things yourself … if someone has already thought it through for you … then it becomes much easier." - Participant 4*

These citations indicate that the prompt analyzer affects ease of use primarily by transforming the act of prompting from an individual cognitive task into a guided one. Rather than requiring users to draw on prompting expertise they may not feel they possess, the prototype offers a structured starting point from which users can refine their input through selection instead of composition.

4.3.4 Usefulness. Usefulness was another factor that participants frequently connected to the prompt analyzer. The feature was generally seen as increasing the practical value of the spend agent, particularly in situations characterized by unfamiliarity, exploration, or uncertainty about what the agent required. One participant described how the analyzer would make the agent more useful from the start, particularly in situations of unfamiliarity:

> *"I think it would be more useful from the start, especially in situations where I don't know the agent or don't know what's needed." - Participant 2*

Another participant highlighted the role of the analyzer in preventing users from giving up early, a pattern that had been reported in the initial interviews, where users stopped trying after a few unsuccessful attempts:

> *"If you try to write it yourself and don't succeed after three or four attempts, you might give up, but if it suggests something, you can move forward using those suggestions and get more out of the agent." - Participant 3*

Taken together, these reflections suggest that the prompt analyzer increases the perceived value of the agent, both immediately, by making it more useful from the start, and over time, by helping users continue rather than give up after unsuccessful attempts.

4.3.5 Capability. Effects on perceived capability were more mixed. Some participants felt that the analyzer changed how they thought about what the AI system was able to do, while others made a clear distinction between the system asking good questions and the system actually understanding the underlying problem. One participant expressed this distinction explicitly, comparing the analyzer's behavior to interactions with people:

"I don't know if it changes my perception of what it's capable of. It's like with people - you can get very clever questions back, but there can still be an equally empty understanding of the problem." - Participant 1

Another participant, however, described the analyzer as a way of making the agent's requirements and possibilities more visible, which in turn shaped their understanding of what the system could realistically do:

"But also, if you have prompt analyzers like this, you understand a bit more about what the Spend Agent might be capable of, or what it needs to function well." - Participant 4

This means that the prompt analyzer affects participants' view of capability less directly than the other UX factors. For some users, it shapes their perception of what the agent can do but for others, the ability to ask good questions is not taken as a sign of deeper understanding.

5 DISCUSSION

This study set out to identify which UX factors are most important when designing to support the adoption of LLM-based AI systems among domain experts with limited AI knowledge in an enterprise environment. The work combined a literature review, semi-structured interviews with sourcing professionals at Ericsson, a prototyping phase consisting of the design of a prompt analyzer and a hands-on evaluation. Five UX factors emerged as most important; reliability, transparency, ease of use, usefulness and a fifth factor empirically identified in this study, perceived capability. These factors also informed the prototyping phase, in which the prompt analyzer was designed and evaluated as a concrete attempt to support several of the identified factors.

5.1 Implications

The literature review identified a broad range of UX factors relevant to technology and AI adoption. While many of these factors are grounded in general technology adoption theory, certain factors appear particularly important in the context of AI specifically. This may be explained by the nature of AI interaction itself: when users delegate cognitive work and decision-making to a system, the stakes of that delegation are high. Unlike traditional software tools, LLM-based systems do not simply execute predefined instructions - they interpret, infer, and generate. This means users must not only understand how to operate the system, but also evaluate whether its outputs can be trusted.

This dynamic is reflected in the empirical findings of this study. Participants working in a high-stakes professional context, where accuracy in figures, contracts, and data analysis carries significant consequences, demonstrated a clearly skeptical attitude toward the AI system. In the enterprise environment studied at Ericsson, users were cautious and critical to rely on outputs that they could not independently verify. In high-precision domains, where even a single error can have cascading consequences, such skepticism reflects not a failure of adoption but a rational response to uncertainty. It is therefore not surprising that reliability and transparency emerged as two of the most important UX factors in this context. When users cannot afford to take outputs for granted, the consistency and accuracy of the system's responses become a baseline requirement rather than a desirable feature. Perceived reliability functions as a kind of threshold: below it, users disengage entirely, and above it, continued engagement becomes possible. This was clearly illustrated by participants who described abandoning the system after a single incorrect result, suggesting that trust in this context is fragile and difficult to rebuild once broken. For AI adoption to take hold in environments like sourcing at Ericsson, systems must therefore demonstrate

a level of reliability that matches the precision standards users already apply to their own professional work.

Transparency serves a complementary but distinct role. Where reliability addresses whether outputs can be trusted, transparency addresses whether users can evaluate that trustworthiness for themselves. In the absence of transparency, users are left to either accept or reject outputs without the means to reason about them critically, a situation that participants in this study found deeply problematic. The ability to trace how a conclusion was reached, verify the sources behind a recommendation, and understand what assumptions the system made are not merely helpful features. In a professional context where accountability and informed decision-making are central, they are prerequisites for meaningful engagement. Designing AI systems that make their reasoning visible and their sources accessible is therefore not only a matter of user experience but of enabling the kind of critical human oversight that high-stakes work requires.

A further factor that emerged from the empirical data, and one not explicitly foregrounded in the existing literature, was perceived capability. Participants frequently reasoned about the AI system in terms of its current maturity, questioning whether it had developed to a point where it could be meaningfully relied upon for professional work. These beliefs functioned as a kind of frame through which system performance was interpreted: when something went wrong, users often attributed it to the immaturity of the technology rather than to a specific interaction failure. This has important implications for adoption, as users who perceive the system as underdeveloped may restrict their use to low-stakes tasks or disengage altogether, regardless of actual performance. Organizations seeking to increase adoption must therefore actively work to build realistic and accurate understandings of AI capability among their employees. This cannot be achieved through abstract communication alone. It requires concrete, relatable demonstrations, ideally shared by a colleague in a familiar work context, that make the practical potential of the system visible and credible.

Perceived usefulness and ease of use, while more established in the technology adoption literature, proved equally relevant in this context, particularly given the profile of the intended user group. Many participants were experienced professionals who had developed well-functioning ways of working over many years, and while some were eager to experiment, others did not approach the AI system with the same curiosity or openness. For these users, the motivation to engage depends heavily on whether the system can be seen to offer genuine value in relation to tasks they already care about. Usefulness in this sense is not abstract. It is tied to specific work situations, and it often needs to be demonstrated rather than explained. At the same time, even users who recognize potential value may not engage if the interaction itself feels demanding or unfamiliar. Ease of use therefore matters not only as a comfort factor but also as a condition for both initial and continued engagement, particularly among users who lack confidence in prompting and interacting with AI systems. If the cognitive effort required just to use the system is high, those without strong intrinsic motivation will return to established ways of working. This highlights the importance of designing simple and intuitive user experiences in AI systems.

5.2 Designing for Adoption

The exploratory design phase took the identified UX factors as its starting point, grounding design decisions in a concrete understanding of what users struggled with and what they needed in order to engage with the system more effectively. Participants frequently expressed uncertainty about how to formulate effective prompts, and several described abandoning the system after receiving outputs that did not meet

their expectations, without understanding why. Since the quality of an LLM-based system's output is directly dependent on the quality of the input it receives, prompting represents a critical point of friction for users with limited AI experience. This observation motivated the prototyping phase of the study, in which a prompt analyzer was designed as a concrete attempt to address this friction and, in doing so, support several of the UX factors identified as most important for adoption.

The evaluation showed that the prompt analyzer had a positive influence across several of these factors. Perceived reliability was strengthened, as the feature helped users provide the information the agent actually needed rather than leaving important details unspecified, reducing the risk of outputs that did not reflect the user's goal. Transparency was the factor most consistently associated with the prototype. By surfacing what the agent needed before a response was generated, the analyzer made visible aspects of the system's reasoning that would otherwise have remained hidden, directly addressing the black box experience. Ease of use was also positively affected, as the guided and selection-based interaction reduced the cognitive effort of formulating prompts and lowered the threshold for users who felt uncertain about their prompting ability. Then the perceived usefulness of an AI agent also increased, particularly in early interactions with an unfamiliar agent, and in supporting users to continue rather than give up after unsuccessful attempts. Effects on perceived capability were more varied. Some participants felt the analyzer gave them a clearer sense of what the system could realistically do. Others, however, were careful not to equate the ability to ask good clarifying questions with a deeper understanding of the problem at hand.

Taken together, this design approach, of anchoring prototyping decisions to empirically identified UX factors, proved valuable in several aspects. It provided a clear rationale for design choices and enabled meaningful evaluation by ensuring that the prototype was directly connected to real user needs and observed challenges. Rather than designing in abstraction, the process ensured that each design decision could be traced back to a specific adoption barrier identified in the study. In doing so, it strengthened the link between theory, empirical findings, and design practice, and demonstrated how UX factors can be operationalised into concrete design interventions to increase AI adoption in enterprise environments.

5.3 Limitations

This study has several limitations that should be considered when interpreting the findings. First, the empirical material is based on a relatively context-specific sample. The interviews and prototype evaluation were conducted with employees within the sourcing function at Ericsson, which means that the findings are closely tied to a specific organizational setting and type of work. While this provides depth and relevance for the studied context, it also limits the generalizability of the results to other domains, industries, or user groups with different levels of AI exposure or different task characteristics.

The study also focuses specifically on users with limited AI knowledge, which was intentional given the research aim. However, this also means that the findings primarily reflect early-stage interactions with AI systems rather than long-term adoption patterns. As users become more experienced, their perceptions of factors may evolve, potentially shifting the relative importance of the identified UX factors over time. The prototype evaluation was exploratory in nature and conducted with a limited number of participants. Although the qualitative feedback provided valuable insights into how the prompt analyzer influenced user perceptions, the short interaction time and controlled setting may not fully reflect real-world usage. In particular, long-term effects on adoption behavior, workflow integration, and sustained usage could not be assessed within the scope of this study.

Another limitation concerns the thematic analysis process. Since the analysis was conducted by a single researcher, there is a potential risk of interpretive and confirmation bias in how the interview material was categorized and interpreted. While the iterative analysis process and continuous supervision aimed to mitigate this, the findings may still have been influenced by prior expectations shaped by the literature review. In addition, although the UX factors are treated as analytically distinct throughout this study, they are in practice closely interrelated. For instance, reliability and transparency mutually reinforce one another, as do ease of use and usefulness. This separation was a deliberate choice intended to enable clearer insights and more focused evaluation, but it should be noted that users experience these factors as intertwined rather than independent dimensions.

Furthermore, the study adopted an exploratory design approach, allowing the prototype and design process to evolve iteratively based on emerging insights and practical constraints. While the identified UX factors were continuously considered throughout the design process and informed many of the design decisions, these relationships were not systematically documented or explicitly mapped during development. Instead, the connection between specific design features and UX factors was primarily established retrospectively through the evaluation phase. A more structured design rationale process, for example through explicit feature-to-factor mapping, could have strengthened the transparency and reproducibility of the design process by making clearer how individual design decisions were intended to support specific UX factors.

5.4 Future work

Future research could extend this work in several directions. Since this study primarily captured early-stage interactions with AI systems among users with limited AI knowledge, a valuable next step would be to investigate how perceptions of the identified UX factors evolve over time. Longitudinal studies following users throughout a longer adoption process could provide deeper insight into how familiarity, experience, and repeated interaction influence these UX factors. Such work could help clarify whether early skepticism decreases as users gain confidence in the system, or whether certain adoption barriers persist even with continued use.

Future research could also explore the identified UX factors across different organizational contexts, industries, and user groups. Since the findings of this study are closely tied to the sourcing context at Ericsson, comparative studies in environments with different task characteristics, levels of risk, or degrees of AI maturity could help determine which UX factors are broadly applicable and which are more context-dependent. In particular, domains involving more creative or less high-stakes tasks may reveal different relationships between trust-related factors, usability, and adoption behavior.

Another direction for future work concerns the thematic relationships between the identified UX factors. Although the factors were analytically separated in this study to support clearer analysis and evaluation, participants often experienced them as interconnected. Future studies could therefore investigate these relationships more explicitly, for example by examining how transparency influences perceptions of reliability, or how ease of use shapes perceived usefulness over time. Exploring such interdependencies could contribute to a more holistic understanding of AI adoption from a UX perspective.

Finally, the prompt analyzer concept itself also offers several opportunities for further development. One important direction would be to investigate how to define a meaningful stopping point in the refinement process, enabling the system to recognize when sufficient information has been provided rather than

continuing to generate suggestions indefinitely. Developing more adaptive and context-aware prompting support could improve both usability and interaction quality, particularly for users with limited AI experience.

5.5 AI Disclaimer

This thesis is the author's own work, and all ideas, analyses, arguments, and conclusions presented are the author's own contributions. Large Language Models (LLMs), including ChatGPT and Claude using various models, were used solely as supportive tools for language improvement, proofreading, and translation assistance.

6 CONCLUSION

This study set out to investigate which UX factors are most important for supporting the adoption of LLM-based AI systems among domain experts with limited AI knowledge in an enterprise environment. Through a literature review, semi-structured interviews, prototype exploration and evaluation, five key factors were identified: reliability, transparency, ease of use, usefulness, and perceived capability.

The findings suggest that AI adoption in enterprise contexts cannot be understood solely in terms of system performance. For domain experts who combine high professional standards with limited AI experience, adoption depends on whether the system can be trusted, understood, and effectively integrated into their work practices. In high-stakes environments such as sourcing at Ericsson, where accuracy and accountability are critical, reliability and transparency emerged as foundational requirements rather than desirable additions. Without these, users were quick to disengage and hesitant to re-engage. The empirically identified factor of perceived capability adds an additional dimension not fully captured in prior literature, highlighting how users' beliefs about the maturity of AI technology shape their willingness to rely on it. Meanwhile, usefulness and ease of use, which are well established in the technology adoption literature, remained highly relevant, particularly in ensuring that users could see clear value in relation to their tasks and interact with the system without high effort or uncertainty.

The prototyping phase demonstrated that these UX factors can be actively addressed through design. By anchoring design decisions in empirically identified factors, each choice could be traced back to a specific adoption barrier observed in the study, providing a well-grounded basis for both development and evaluation. The prompt analyzer was developed as a response to the prompting difficulties observed in the interviews. The results showed positive effects across several of the identified UX factors, suggesting that the solution can support AI adoption among domain experts with limited AI experience in enterprise environments.

As this study was conducted within a single organizational context, future research should explore how the identified factors apply across a broader range of settings and user groups, and examine how both adoption patterns and the relative importance of specific UX factors may shift as users gain more experience with AI systems over time.

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